

Getting Started with Conda Environments



Why Should I Care About Conda?

🧠 Imagine working on two different science projects.

- One needs **Python 3.10** and Package A
- The other needs **Python 3.8** and Package B

Running both on the same system can cause **conflicts**.

```
#~~~~~#  
# Conda helps you avoid that. #  
#~~~~~#
```

🎒 Conda is like a backpack for each project.

- You pack only what you need
- Keep it separate from other projects
- You can “activate” or “switch” between backpacks anytime



What is Conda?

- Conda is a tool that helps install software and manage environments
- It's like an organizer for your digital workspace (especially when you working on local device or workstations).
- It avoids the headache of software crashes due to mismatched versions.



Benefits:

- No coding errors due to conflicting setups.
- Easy to try out new tools without “breaking” your computer (local device/ workstation/HPCs).



Think of a House...



Imagine a house with two separate rooms.

- In Room A, the person is very neat – everything is clean and organized.
- In Room B, the person is very messy – clothes on the floor, food everywhere.

Even though both people live under the same **roof (your system)**, they don't bother each other because they are in **separate rooms (environments)**.



The tidy person keeps their space clean.



The messy person does what they want in their space.

Result: No clashes, the rest of the house remains peaceful.



Imagine There Are No Rooms...

 What if both people had to live in one big shared space – no walls, no doors?

- One keeps cleaning.
- The other keeps making a mess.
- Soon, they start fighting, arguing, and messing up each other's things.

 **This is exactly what happens when you don't use environments in programming:**
Conflicts, errors, broken tools.



Conda Environments Keep Things Separate and Safe

Just like rooms in the house, conda isolates each project into its own clean, separate space:

- You can install specific tools in each room
- Different room can have different versions of Python
- What you do in one room never affects the other.

 **It's like giving each person their own backpack, space, and freedom — and your house (computer) stays healthy and sane!**



Installing Conda (Windows)

We'll use **Miniconda: a lightweight version of Anaconda*

For Windows:

1. Open your Command Prompt (Search cmd)
2. Type the following commands:

There are three separate commands:

- **First command**, downloads the miniconda software installer from the internet.
- **Second command**, starts the installation using the downloaded installer.
- **Third command**, deletes the installer since the miniconda is already installed

```
curl https://repo.anaconda.com/miniconda/Miniconda3-latest-Windows-x86\_64.exe -o .\miniconda.exe
```

```
start /wait "" .\miniconda.exe /S
```

```
del .\miniconda.exe
```



Installing Conda (MacOS)

Same as windows we will be downloading and installing miniconda using the mac terminal.

- Since the mac comes in two different kind of chipset (Intel and Apple Silicon) the installation is a bit different.
- You can check if your mac has intel chip or Apple silicon chip (M1, M2 or M3) and use the commands accordingly.

```
mkdir -p ~/miniconda3  
curl https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-x86_64.sh -o ~/miniconda3/miniconda.sh  
bash ~/miniconda3/miniconda.sh -b -u -p ~/miniconda3  
rm ~/miniconda3/miniconda.sh
```

For Intel based mac

For Apple Silicon
mac

```
mkdir -p ~/miniconda3  
curl https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-arm64.sh -o ~/miniconda3/miniconda.sh  
bash ~/miniconda3/miniconda.sh -b -u -p ~/miniconda3  
rm ~/miniconda3/miniconda.sh
```



Installing Conda (MacOS)

Once installation is complete:

1. Restart your command prompt (if Windows) or terminal (if Mac)
2. Check if the conda is correctly installed by typing the following command:

```
conda --version
```

- This will print the version of conda installed in your device. Once confirmed run:

```
conda activate
```

- This command will start the conda base environment(This is your **House!**)

```
[samaksh@Samakshs-MacBook-Pro ~ % conda --version  
conda 24.5.0  
[samaksh@Samakshs-MacBook-Pro ~ % conda activate  
(base) samaksh@Samakshs-MacBook-Pro ~ % █
```

Once you see the (base); It means that now you are in the **House!**

- This environment is also called the Global environment.



Creating New Room

If you remember from our story; we need to have **rooms (local environments)** in the **house(global environment)** to prevent clashes between the housemates(**Python/Softwares/Tools**).

- **Now to create a local environment we use the following command**

```
conda create --name myenv python=3.11
```

*At some point of creating the conda environment you will be asked to press **[y]/[n]**, Just type 'y' and press enter.*



Creating New Room

Once the installation is complete type the following command:

```
conda activate myenv
```

```
[(base) samaksh@Samakshs-MacBook-Pro ~ % conda activate myenv  
(myenv) samaksh@Samakshs-MacBook-Pro ~ % █
```

Now you can see that we see myenv instead of base!

Congratulations! Now in our house **base**, we have successfully created a room **myenv** !!!

To exit the local environment, type the following command:

```
conda deactivate myenv
```

```
[(myenv) samaksh@Samakshs-MacBook-Pro ~ % conda deactivate  
(base) samaksh@Samakshs-MacBook-Pro ~ % █
```

This will bring you out from your local environment (myenv) back to the global environment (base)



Congratulations!!!

What we have learned so far:

- Installation of miniconda.
- Setting up conda base environment (Our house)
- Creating new local environment (Our rooms)
- Activating and deactivating local environments.

